Report: Multiple Alignment of Geminivirus Capsid Proteins

CodeAlpha Internship, Bioinformatics, Task 2

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1. Objective

My goal is to compare 9 geminivirus capsid proteins. I am using a multiple sequence alignment to identify conserved regions and determine which viruses are most similar to one another.

2. Data and Method

2.1 Sequences

I downloaded 9 capsid protein sequences from UniProt. They range from 244 to 258 amino acids in length and originate from 9 viruses belonging to the *Geminiviridae* family.

Table 1: sequences used

Virus	UniProt	Length (AA)
Tomato latent virus	A0A0B5JFN7	258
African cassava mosaic virus (Ouganda)	O12282	258
African cassava mosaic virus (Kenya)	P03561	258
Cotton leaf curl Gezira virus	Q8V6I9	258
Beet curly top virus	P14984	254
Sweet potato leaf curl virus	Q9YPT1	254
Cabbage leaf curl virus	Q96701	251
Panicum streak virus	Q00323	247
Maize streak virus	B2BX32	244

2.2 Alignment

I aligned the 9 sequences using Clustal Omega, on the EBI website, on October 19, 2026. I kept the default settings.

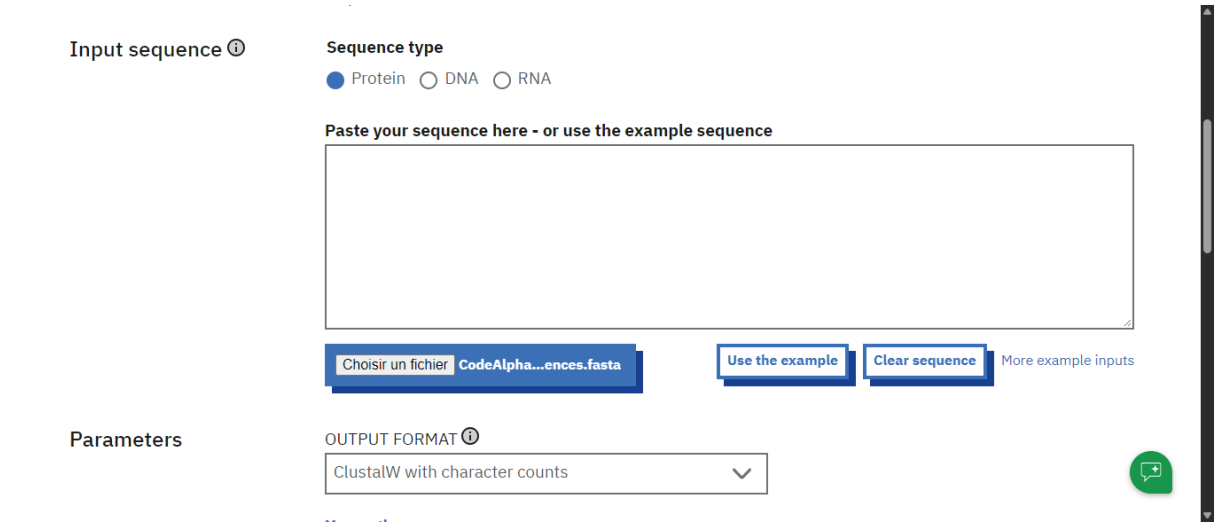


Figure 1: Clustal Omega submission page with the 9 sequences.

3. Results

3.1 General overview of the alignment

The alignment spans 286 positions. It is longer than the sequences themselves because the tool inserts gaps (dashes) to align the proteins. The 6 begomoviruses are similar across nearly their entire length. The other 3 viruses contain large, distinct regions, particularly at the beginning of the protein.

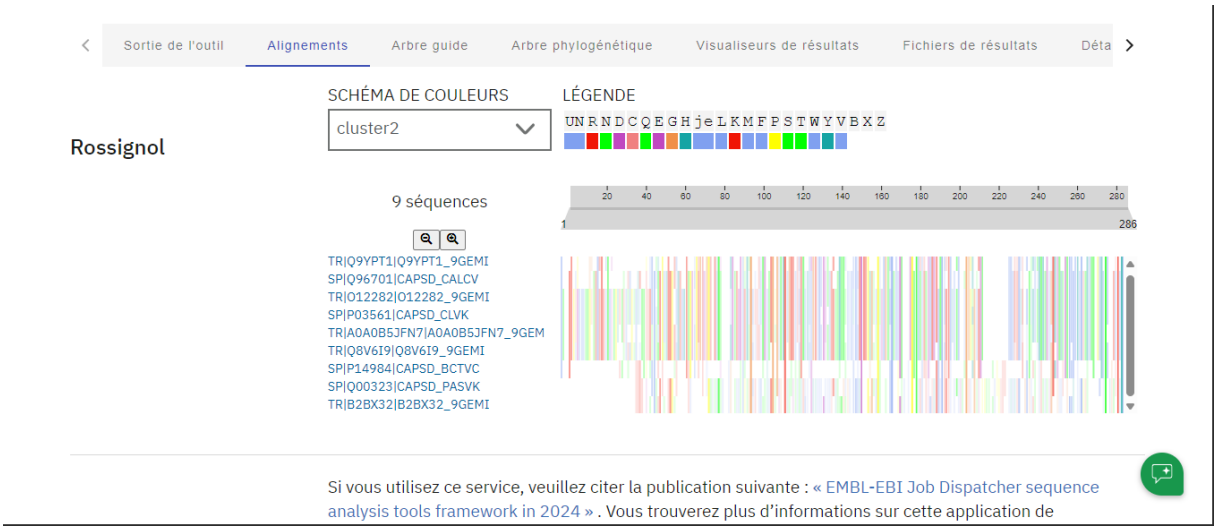


Figure 2: Color-coded alignment of the 9 capsid proteins (Jalview, "Percentage Identity" coloring)

3.2 Conserved regions

Across the 9 viruses: Only 14 out of 286 positions are identical across all 9 viruses, representing approximately 4.9%. These are positions 95, 106, 112, 150, 158, 165, 175, 177, 190, 202, 271, 277, 279, and 280. They are isolated; there are no complete blocks that are identical across all 9 viruses.

Among the 4 closely related viruses: The 4 closely related viruses are the two African cassava mosaic virus (ACMV) isolates, Tomato latent virus, and Cotton leaf curl Gezira virus. In these, 189 positions are identical, representing approximately 66%. The longest conserved blocks are:

- positions 9 to 25, at the beginning of the protein (with only one difference, at position 16)
- positions 59 to 66
- positions 134 to 141
- positions 149 to 155
- positions 247 to 265
- positions 275 to 283, at the end of the protein

3.3 Similarity between the viruses

The identity matrix provides the percentage of similarity between each pair of viruses.

Table 2: Summary of the identity matrix

Comparison	**Identity**
ACMV Ouganda et ACMV Kenya	97,7 %
Tomato latent virus et Cotton leaf curl Gezira	84,1 %
Entre les 4 virus proches	79 à 98 %
Cabbage leaf curl et les 4 virus proches	70 à 73 %
Sweet potato leaf curl et les autres begomovirus	50 à 52 %
Panicum streak et Maize streak	67,5 %
Beet curly top et les autres virus	21 à 25 %
Panicum et Maize streak avec les begomovirus	17 à 20 %

The closest pair is that of the two ACMVs. The most distant pair consists of Panicum streak virus and Cotton leaf curl Gezira virus (16.7%).

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#
#
#  Percent Identity  Matrix - created by Clustal2.1
#
#

     1: tr|Q9YPT1|Q9YPT1_9GEMI          100.00   49.80   51.79   51.39   51.00   49.80   22.52   18.52   17.76
     2: sp|Q96701|CAPSD_CALCV            49.80  100.00   69.72   69.72   72.91   69.72   22.67   18.10   18.35
     3: tr|O12282|O12282_9GEMI           51.79   69.72  100.00   97.67   81.78   80.23   24.12   19.82   20.09
     4: sp|P03561|CAPSD_CLVK             51.39   69.72   97.67  100.00   80.62   79.07   24.56   19.82   20.09
     5: tr|A0A0B5JFN7|A0A0B5JFN7_9GEMI   51.00   72.91   81.78   80.62  100.00   84.11   21.49   17.12   18.72
     6: tr|Q8V6I9|Q8V6I9_9GEMI           49.80   69.72   80.23   79.07   84.11  100.00   21.05   16.67   17.35
     7: sp|P14984|CAPSD_BCTVC            22.52   22.67   24.12   24.56   21.49   21.05  100.00   24.27   24.47
     8: sp|Q00323|CAPSD_PASVK            18.52   18.10   19.82   19.82   17.12   16.67   24.27  100.00   67.49
     9: tr|B2BX32|B2BX32_9GEMI           17.76   18.35   20.09   20.09   18.72   17.35   24.47   67.49  100.00
```

Figure 3: Identity matrix (in %) for the 9 proteins.

3.4 Tree

The tree groups together viruses that are similar to one another. It shows:

- the two ACMV isolates together, with the shortest branches
- Tomato latent virus and Cotton leaf curl Gezira together
- Panicum streak and Maize streak together
- Beet curly top with a very long branch, indicating that it is the most distinct

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(
(
(
tr|Q9YPT1|Q9YPT1_9GEMI:0.24805,
(
sp|P14984|CAPSD_BCTVC:0.35714,
(
sp|Q00323|CAPSD_PASVK:0.16435,
tr|B2BX32|B2BX32_9GEMI:0.16075)
:0.23661)
:0.17050)
:0.10224,
sp|Q96701|CAPSD_CALCV:0.14821)
:0.04576,
(
tr|O12282|O12282_9GEMI:0.00861,
sp|P03561|CAPSD_CLVK:0.01465)
:0.08525,
(
tr|A0A0B5JFN7|A0A0B5JFN7_9GEMI:0.07043,
tr|Q8V6I9|Q8V6I9_9GEMI:0.08849)
:0.01940);
```

Figure 4: Tree of the 9 capsid proteins.

4. Interpretation

The capsid protein is highly conserved among closely related begomoviruses. This conservation decreases significantly when other genera are included. Panicum streak virus and Maize streak virus (mastreviruses) are closely related to each other, but very distant from

the begomoviruses. Beet curly top virus (curtovirus) is also distant from all the others. These groupings align with the established classification of these viruses.

A conserved region is often important for the protein's function. A change in such a region is likely to disrupt that function.

The N-terminal portion of the protein (positions 9 to 25) is rich in lysine and arginine. These amino acids may facilitate DNA binding.

5. Conclusion

The alignment of 9 capsid proteins reveals a high degree of conservation among closely related begomoviruses (189 identical positions), but low conservation across all the viruses examined (14 positions). The viruses cluster according to their evolutionary relationships: ACMV isolates are very closely related, while mastreviruses are distant from the begomoviruses.